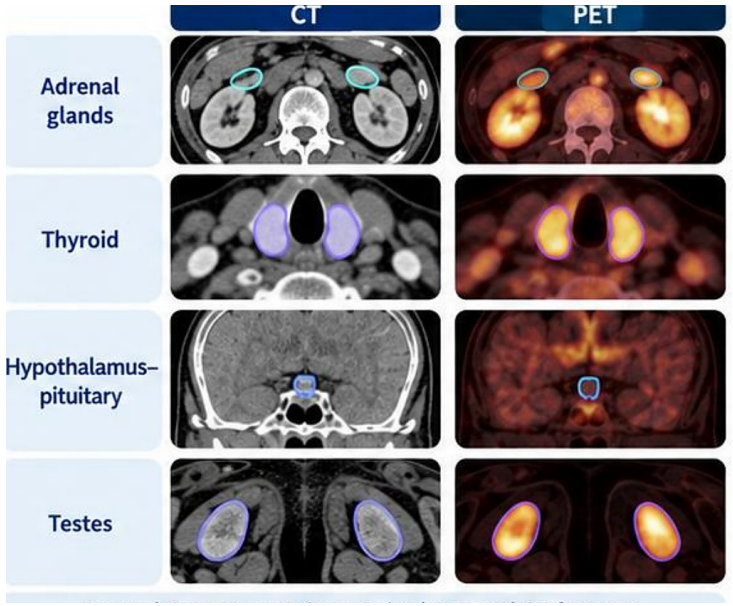
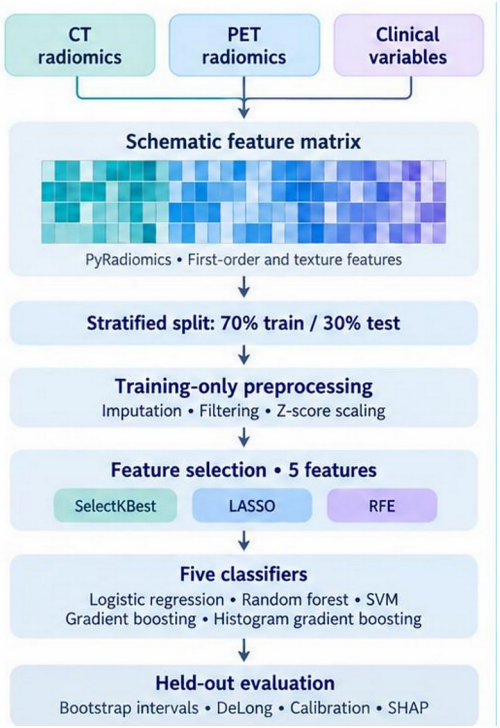
RADIOMICS · SYSTEMIC PHENOTYPING

Endocrine-organ PET/CT radiomics



Endocrine organs on PET/CT



CT + PET + clinical fusion model

Results

Test AUC (held-out 30%)

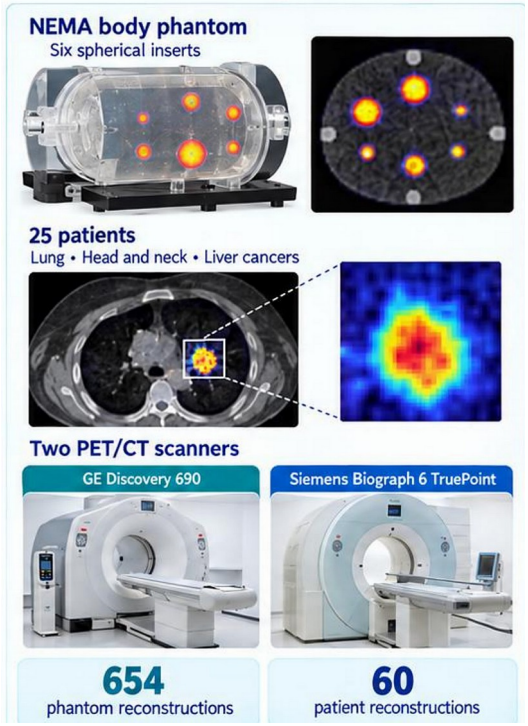
Model	AUC
Clinical only	0.727
CT + clinical	0.729
PET + clinical	0.733
CT + PET + clinical	0.758
95% CI (best)	0.653–0.849
ΔAUC vs clinical	0.031
DeLong p	0.569

Retrospective, single centre

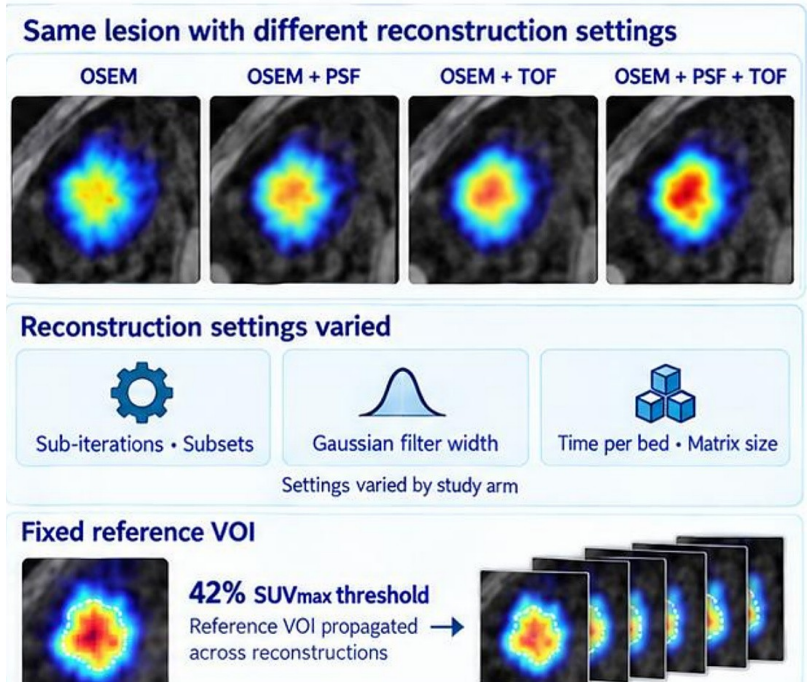
Systemic imaging signatures beyond visible lesions

RADIOMICS · ROBUSTNESS

PET reconstruction changes radiomic features



Phantom + patients, 2 scanners



Same lesion, many reconstructions

Results

Feature variability (COV)

Measure	Features
COV ≤5% (robust)	47%
COV >20% (unstable)	10%
Geometry, COV ≤5%	100%
Intensity, COV ≤5%	44%
Texture, COV ≤5%	41%
Matrix size, COV >20%	56%
Matrix size, COV ≤5%	6%

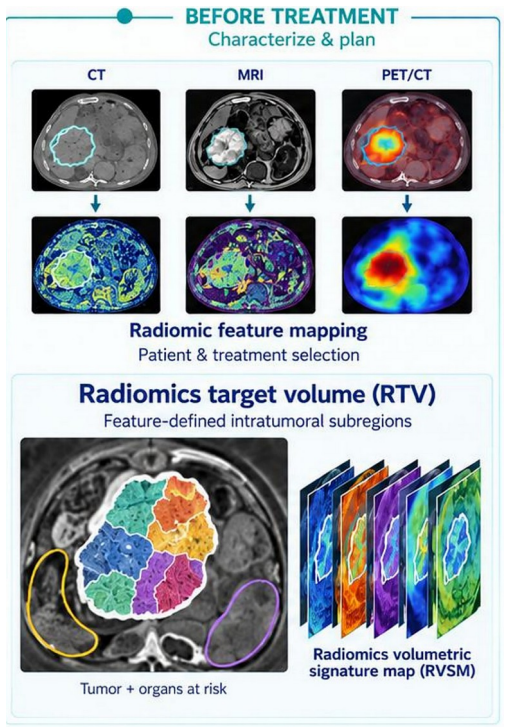
654 phantom + 60 patient reconstructions

Standardize reconstruction before using radiomics

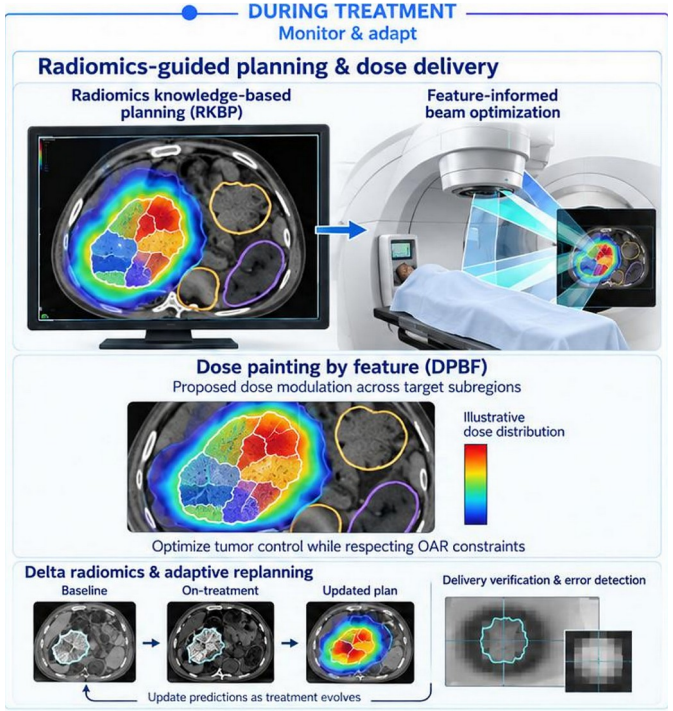
Shiri et al., Eur Radiol 2017

RADIOMICS-GUIDED RADIATION THERAPY

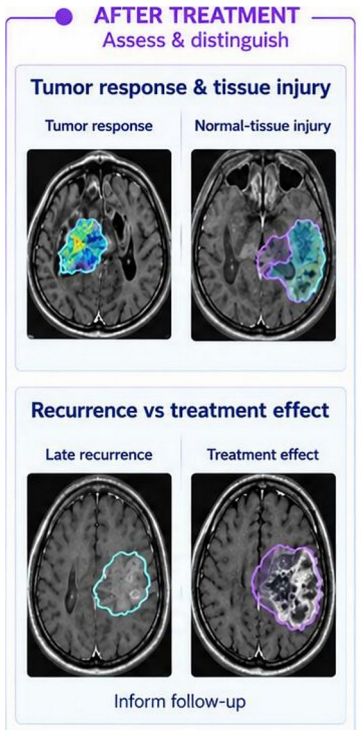
Radiomics across the radiotherapy timeline



Before: characterize and plan



During: monitor and adapt

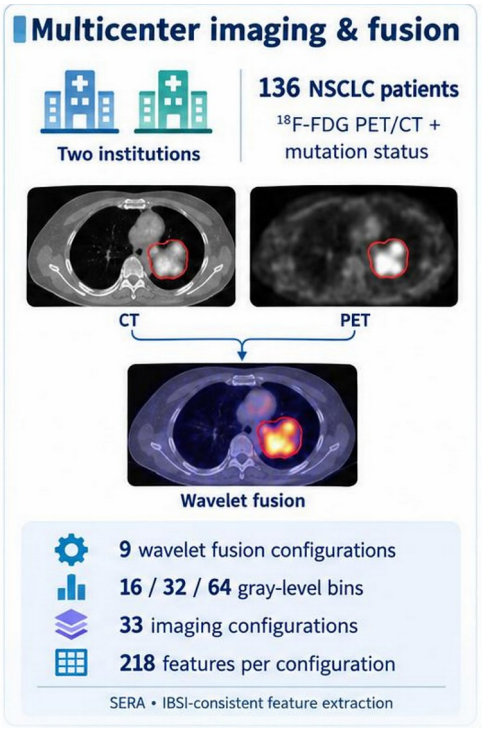


After: assess response

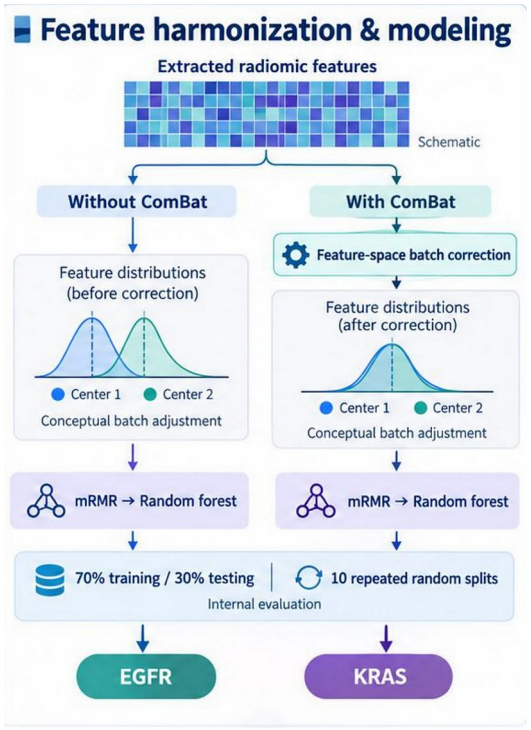
From feature maps to dose painting and adaptive replanning

RADIOGENOMICS · MULTICENTRE HARMONIZATION

ComBat harmonization for EGFR and KRAS



136 NSCLC, two centres



Models with and without ComBat

Results

Mean model AUC

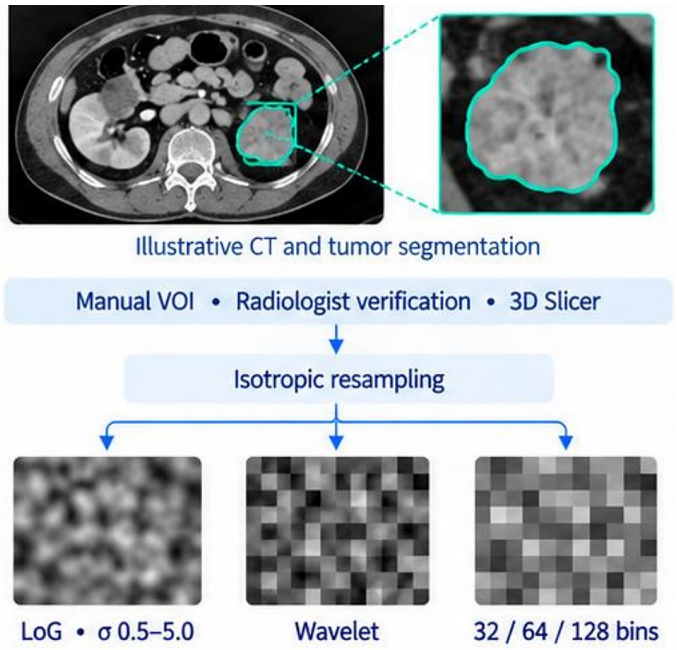
Gene	Before	After
EGFR	0.87–0.90	0.92–0.94
KRAS	0.85–0.90	0.91–0.94
EGFR, best	—	0.94
KRAS, best	—	0.93

Multivariate improvement $q < 0.05$

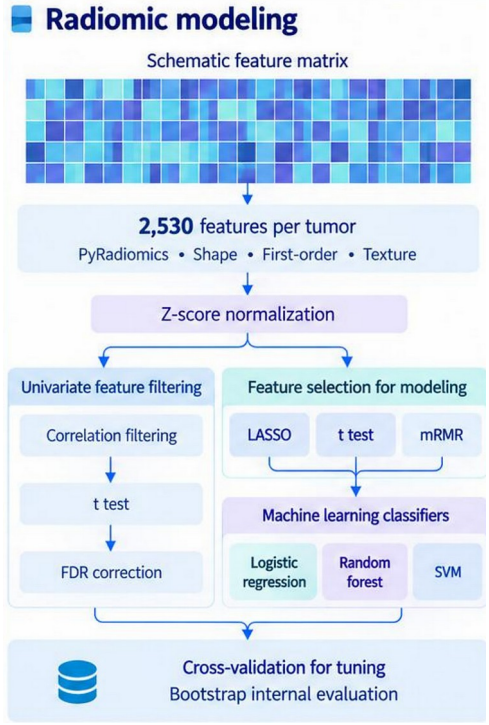
Harmonization makes multicentre radiomics more reliable

RADIOMICS · TUMOUR GRADING

CT radiomics for Fuhrman grade in ccRCC



71 ccRCC CT scans, filtered images



2,530 features → machine learning

Results

AUC by preprocessing

Images	LR	RF	SVM
Original	0.68	0.62	0.76
32 bins	0.73	0.56	0.70
64 bins	0.70	0.55	0.65
128 bins	0.75	0.60	0.77
LoG σ 4.5	0.62	0.62	0.83

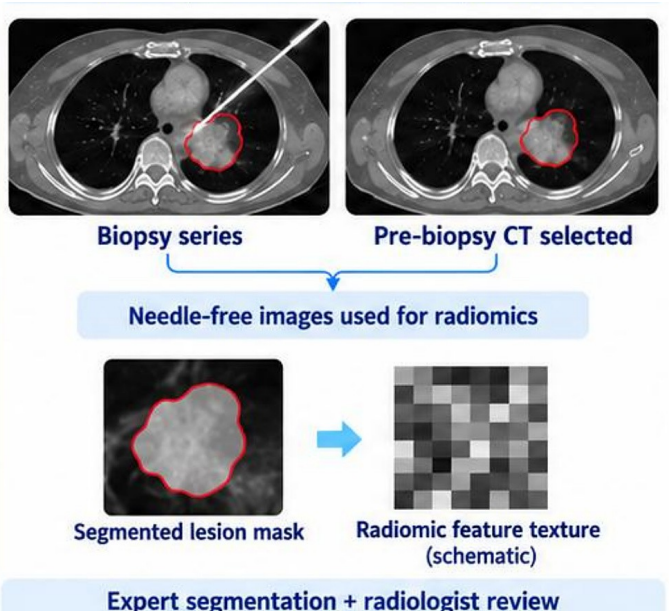
4 texture features significant (FDR q < 0.05)

Potential non-invasive preoperative grading

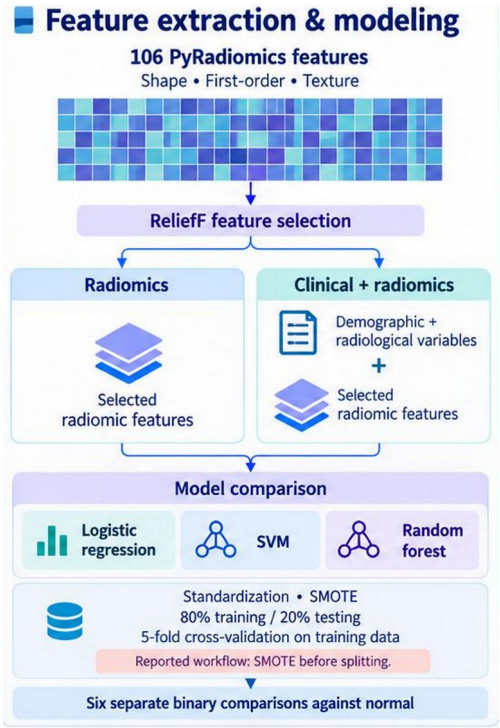
Nazari et al., La Radiologia Medica 2020

RADIOMICS · VIRTUAL BIOPSY

CT-based virtual biopsy of thoracic lesions



Pre-biopsy CT, 300 patients



Feature selection + three classifiers

Results

Test AUC vs normal

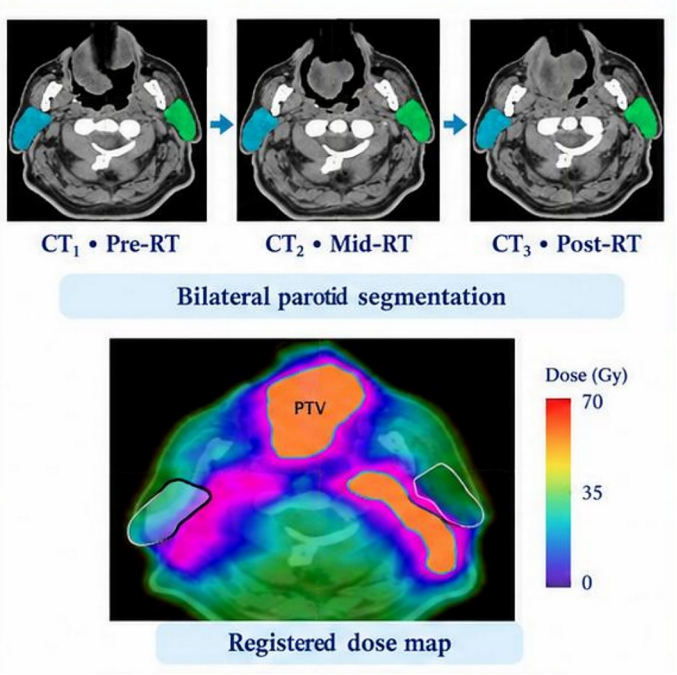
Lesion	Radiom.	+ Clin.
Tumoral	0.682	0.721
Adenocarcinoma	0.690	0.741
SCC	0.820	0.769
SCLC	0.869	0.875
Lymphoma	0.875	0.920
Mesenchymal	0.861	0.852

SMOTE before splitting may inflate results

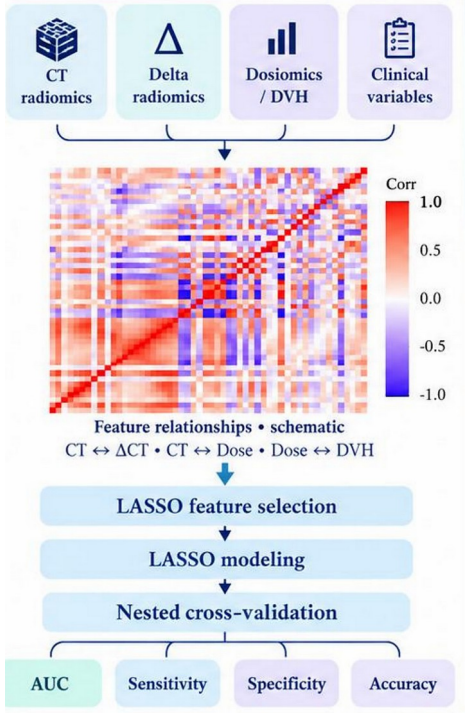
A virtual biopsy alongside tissue diagnosis

RADIOMICS · TOXICITY PREDICTION

Delta radiomics and dose for late xerostomia



Parotid CT before, during, after RT



CT, delta, dose and clinical features

Results

AUC, nested CV

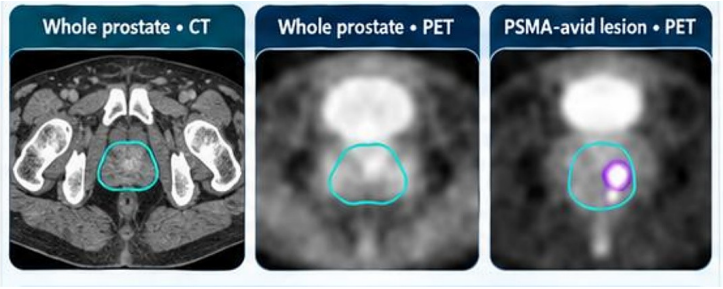
Model	AUC
CT1 (pre-RT)	0.68
CT2 (mid-RT)	0.84
CT3 (post-RT)	0.89
ΔCT2−1 + dose	0.89
DVH	0.89
Dose	0.72
Clinical	0.88

31 patients, 59 parotids

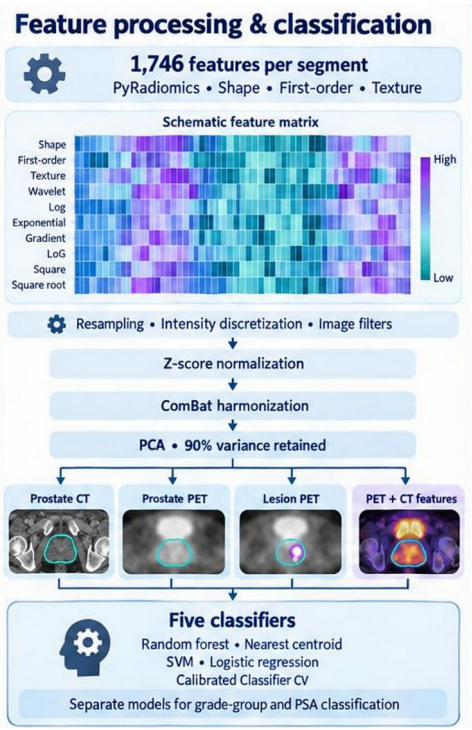
Longitudinal imaging and dose predict dry mouth

RADIOMICS · PSMA PET/CT

Multicentre PSMA PET/CT radiomics for risk



PSMA PET/CT, 243 patients, 4 centres



Harmonized features, five classifiers

Results

Mean accuracy

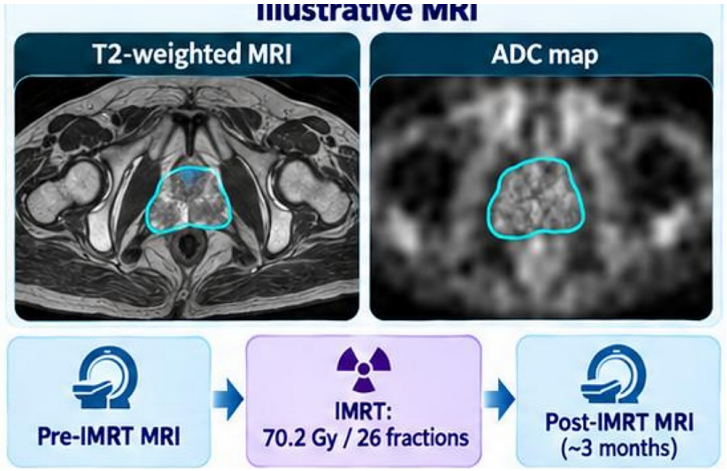
Task	Internal	External
GGG 1–3 vs 4–5	78.0%	73.8%
PSA <20 vs ≥20	75.4%	76.4%

186 development, 57 external test; classifier differences p > 0.05

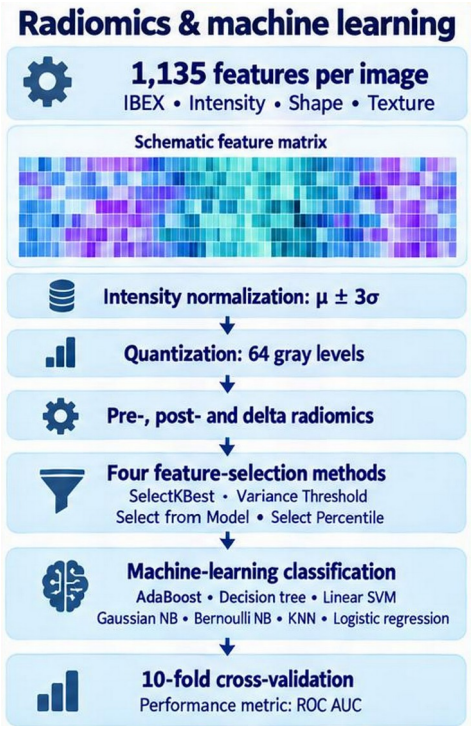
Non-invasive risk stratification from PSMA PET/CT

RADIOMICS · TREATMENT RESPONSE

MRI radiomics for IMRT response in prostate



T2 MRI and ADC before and after IMRT



1,135 features, seven classifiers

Results

AUC

Task	T2w	ADC
IMRT response	0.78	0.77
Gleason	0.739	0.700
Stage	0.625	0.675

Response: best model; Gleason, stage: mean AUC. Responder = ADC rise >20%

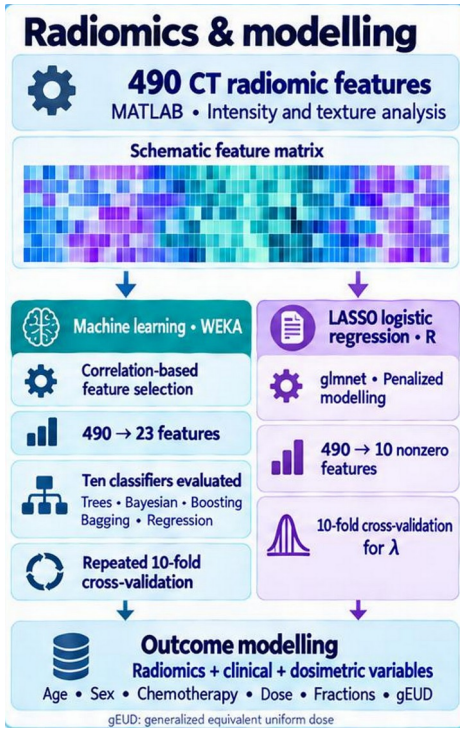
MRI radiomics for response and tumour characterization

COCHLEAR CT RADIOMICS · TOXICITY

Predicting radiation-induced hearing loss



Cochlear CT, 47 patients



490 features, ML and LASSO

Results

Hearing-loss prediction

Model	Result
LASSO radiomics, AUC	0.885
ML accuracy	76.1%
ML precision	75.9%
gEUD only, AUC	0.79
gEUD only, accuracy	72%

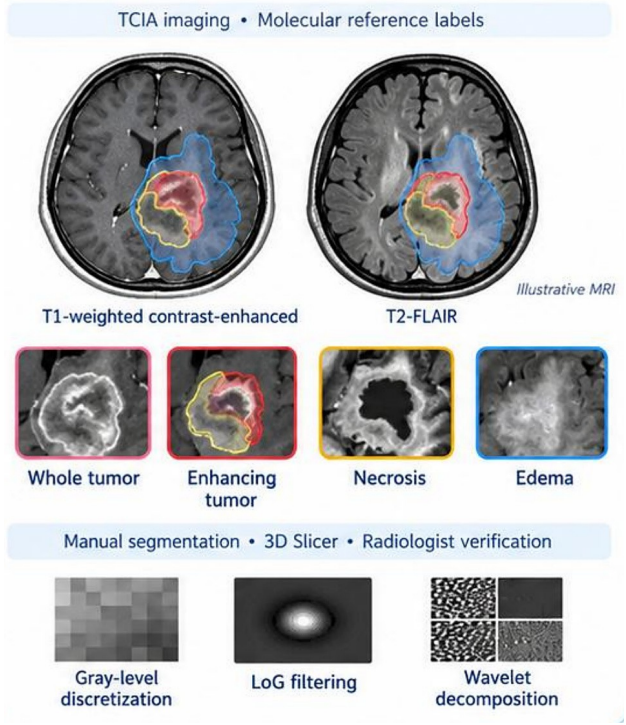
37 of 94 ears with SNHL

Imaging the cochlea to predict hearing loss

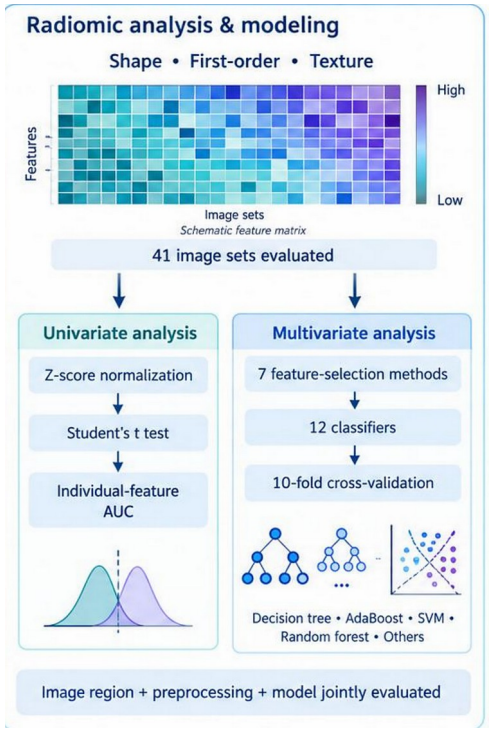
Abdollahi et al., Physica Medica · doi:10.1016/j.ejmp.2017.10.008

RADIOGENOMICS · GLIOBLASTOMA

MRI radiomics for MGMT promoter methylation



MRI tumour sub-regions, 82 patients



41 image sets, 12 classifiers

Results

AUC, 10-fold CV

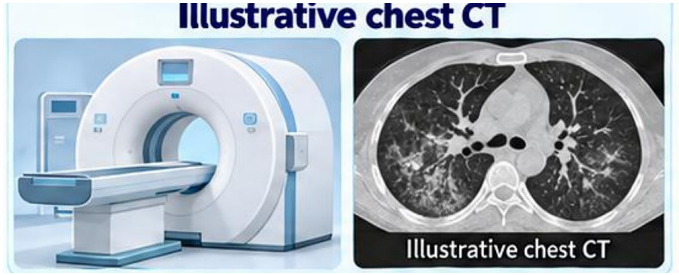
Model	AUC
Decision tree, edema	0.78
AdaBoost, edema	0.74
Best single feature (GLCM)	0.71

45 methylated, 37 unmethylated

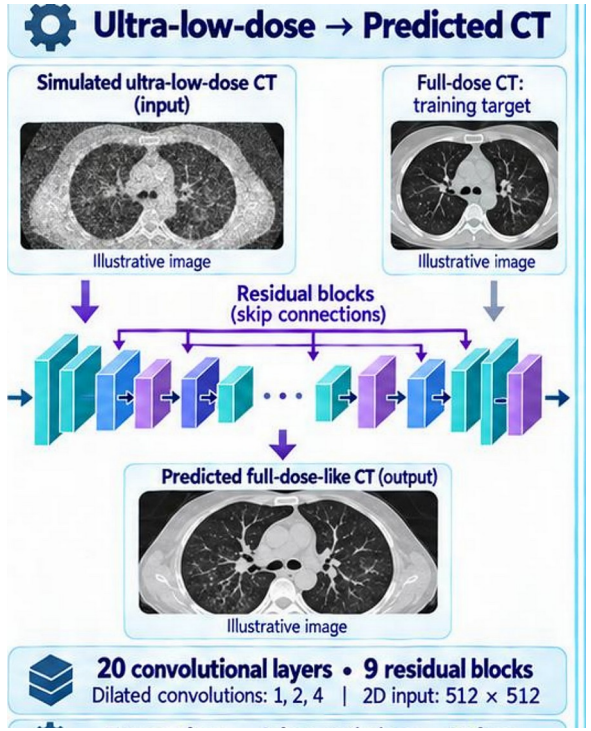
Non-invasive MGMT status from MRI

DEEP LEARNING · IMAGE QUALITY

Deep learning for ultra-low-dose chest CT



1,141 chest CTs, 9 centres



Residual network: low → full dose

Results

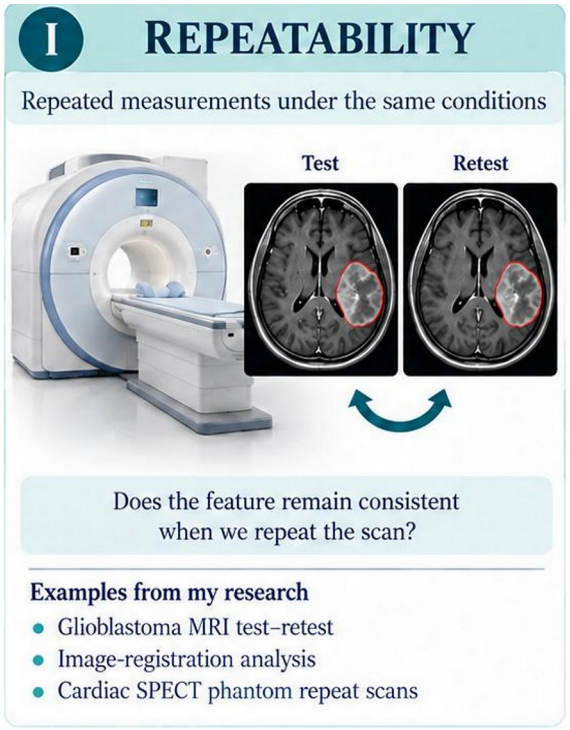
External validation

Metric	ULD CT	DL CT
RMSE	0.16	0.08
SSIM	0.89	0.97
PSNR (dB)	29.40	33.60
Reader score /5	2.78	4.42

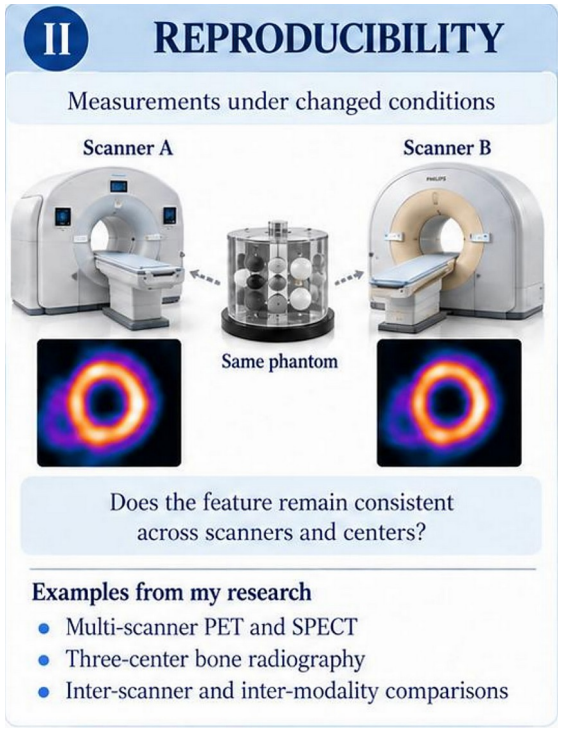
Full-dose reader score 4.72; all $p < 0.0001$

Lower dose with preserved image quality

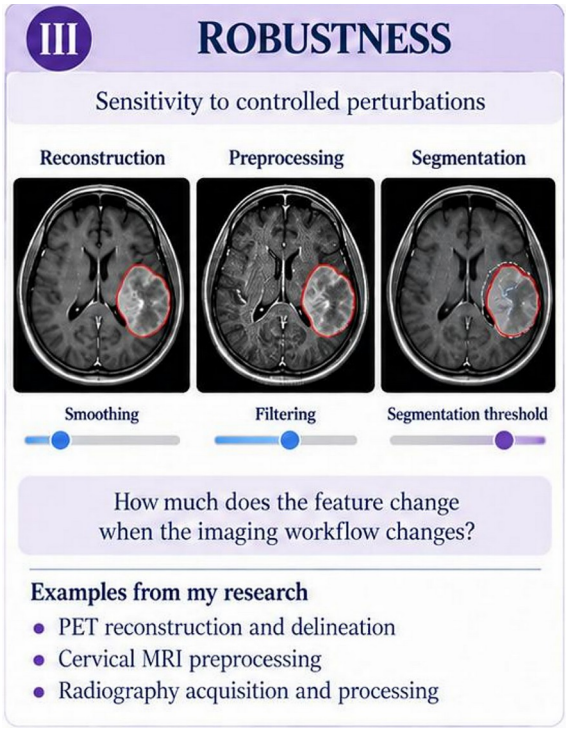
Repeatability, reproducibility and robustness



Repeatability: same conditions



Reproducibility: across scanners

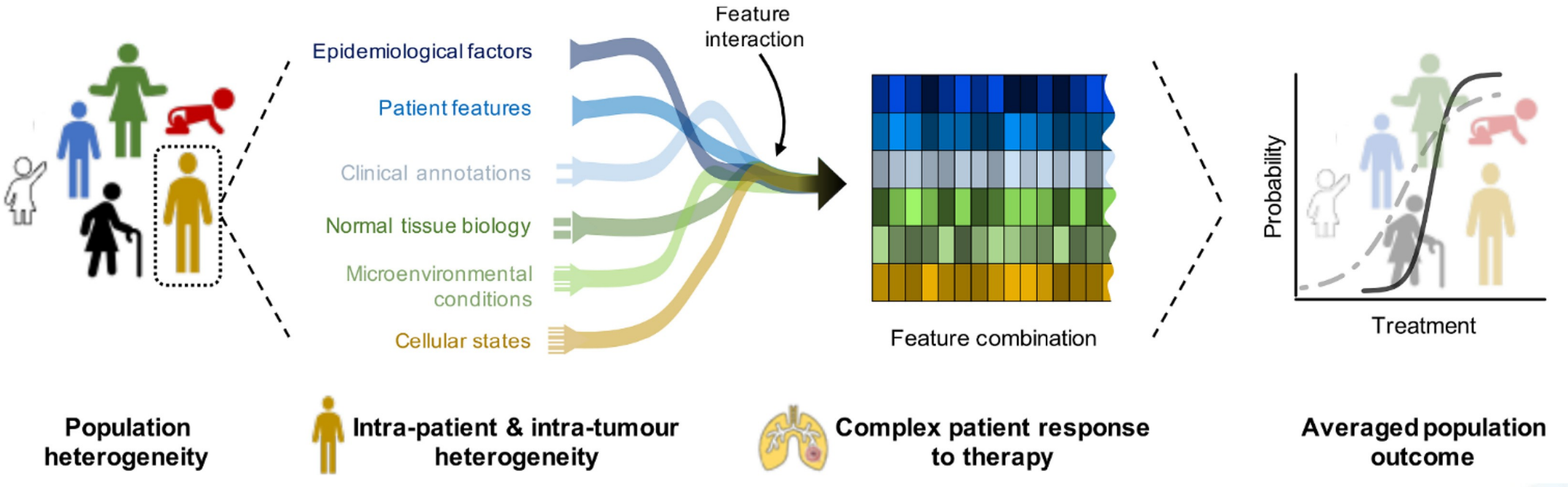


Robustness: workflow changes

Goal: find stable features and the sources of variability

THE CHALLENGE

Treatment is designed for the average patient



Averaging hides the individual patient

THE SOLUTION · DIGITAL TWINNING

Trials for patients who are not average



iScience

CellPress
OPEN ACCESS

Perspective

Designing clinical trials for patients who are not average

Thomas E. Yankeelov,[1,2,3,4,5,9,*] David A. Hormuth II,[4,5] Ernesto A.B.F. Lima,[4,6] Guillermo Lorenzo,[4,10] Chengyue Wu,[4] Lois C. Okereke,[4] Gaiane M. Rauch,[7] Aradhana M. Venkatesan,[7] and Caroline Chung[8]

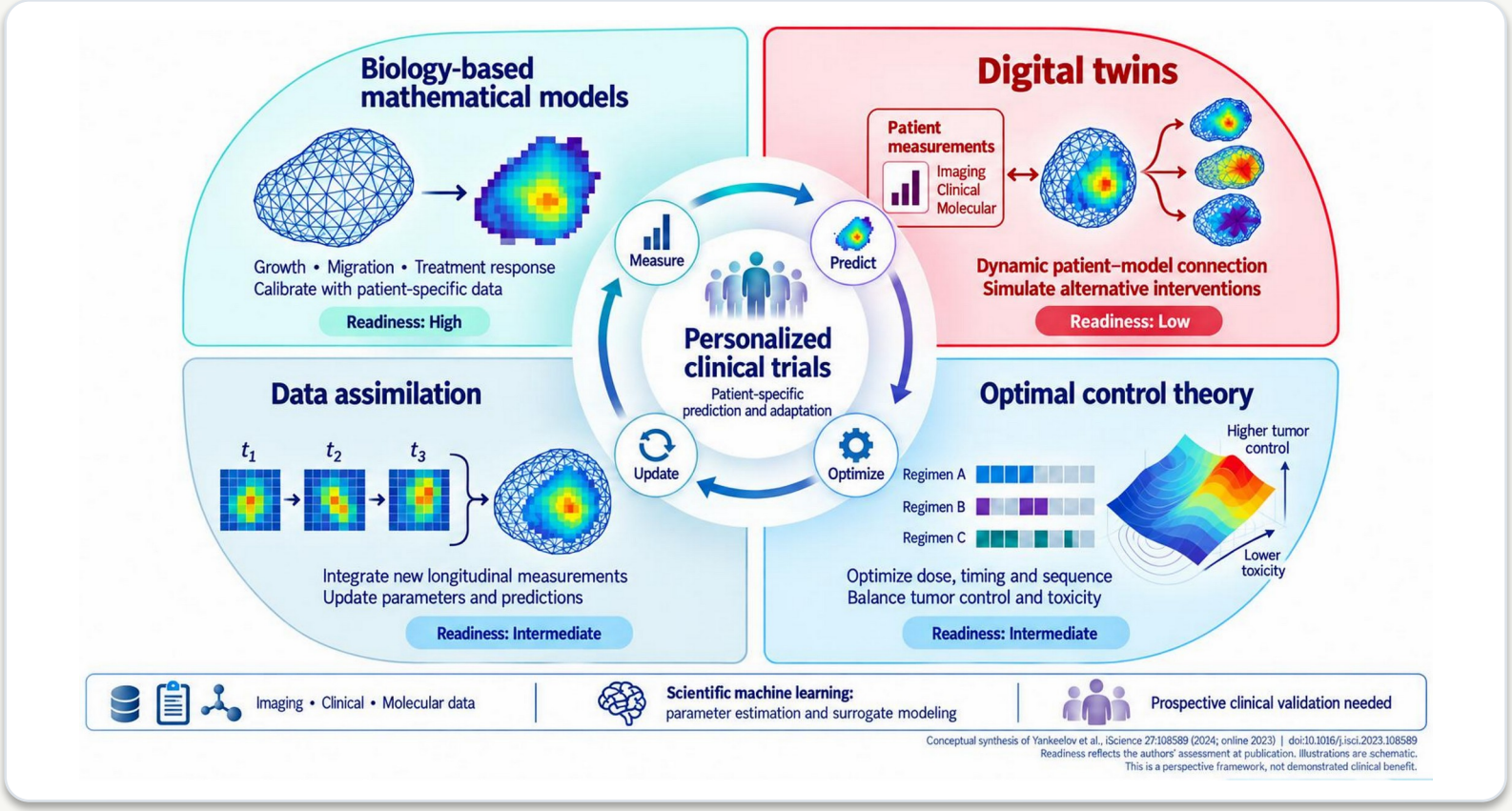
The idea

- Model each patient, not the average
- Calibrate with their own data
- Update as treatment proceeds

From averaged evidence to personalized, adaptive decisions

THE SOLUTION · DIGITAL TWINNING

Four building blocks for personalized trials



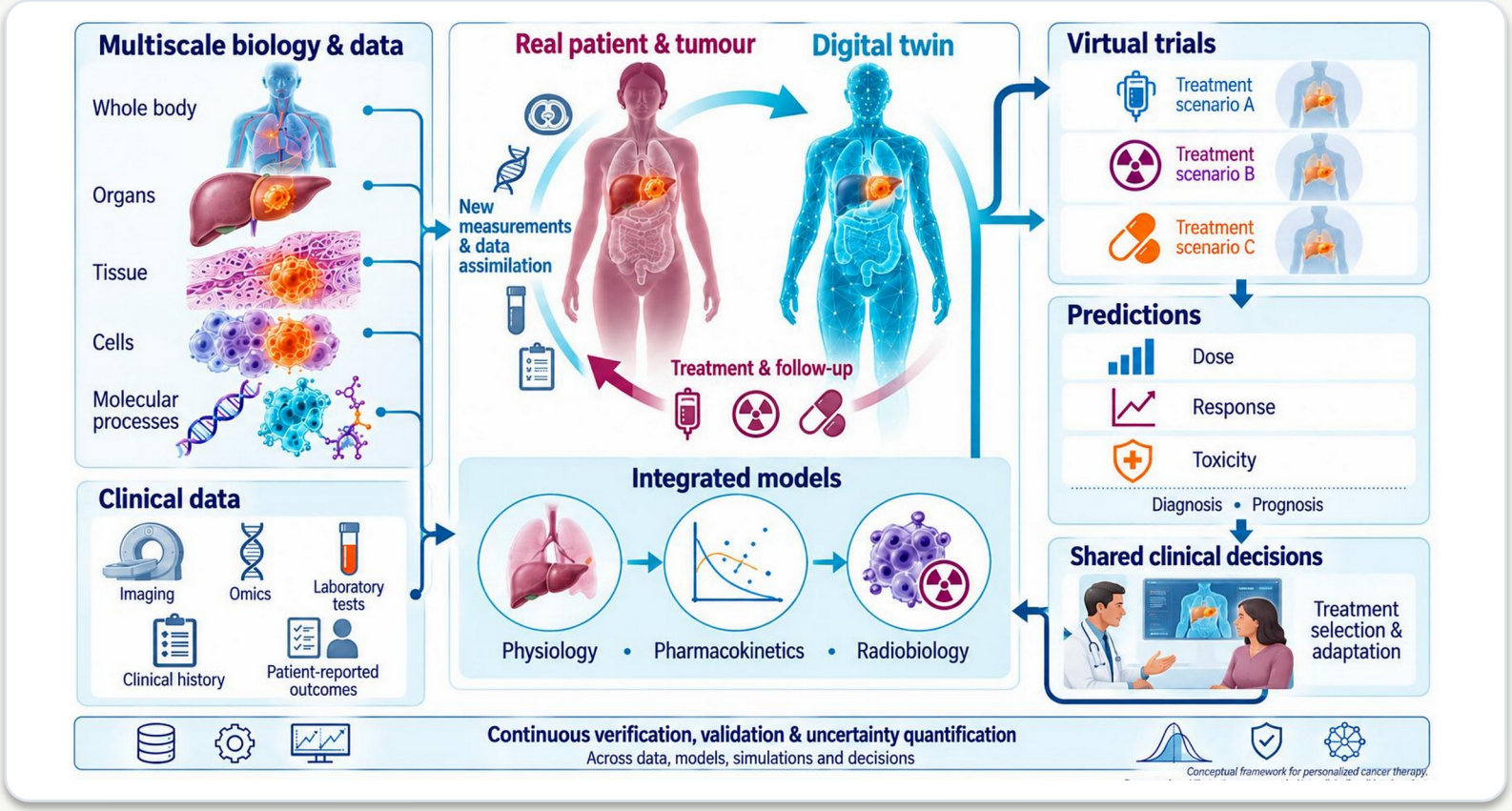
Building blocks

- Biology-based models
- Data assimilation
- Optimal control
- Digital twins

Measure → predict → optimize → update

THE SOLUTION · DIGITAL TWINNING

What is a digital twin?



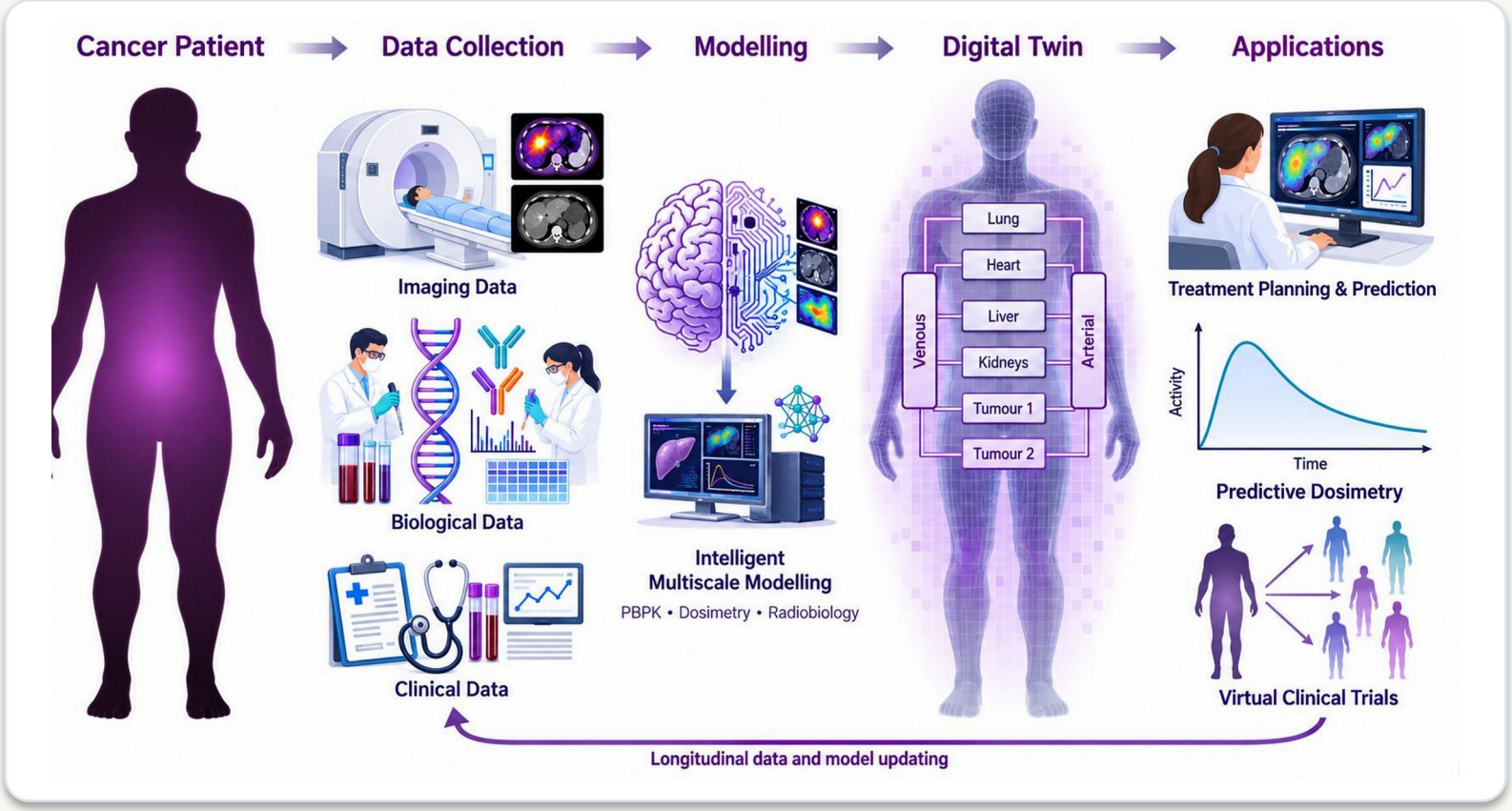
Definition

- A virtual copy of one patient
- Updated with that patient's data
- Predicts and guides decisions

A two-way link between patient and model

THERANOSTIC DIGITAL TWINNING

From the patient to a theranostic digital twin



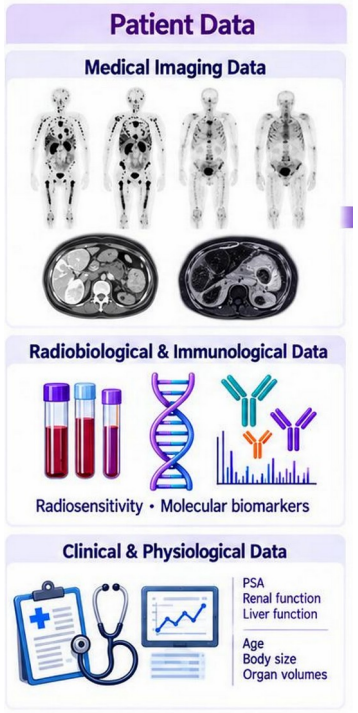
Steps

- Imaging, biological, clinical data
- PBPK, dosimetry, radiobiology
- Planning and virtual trials

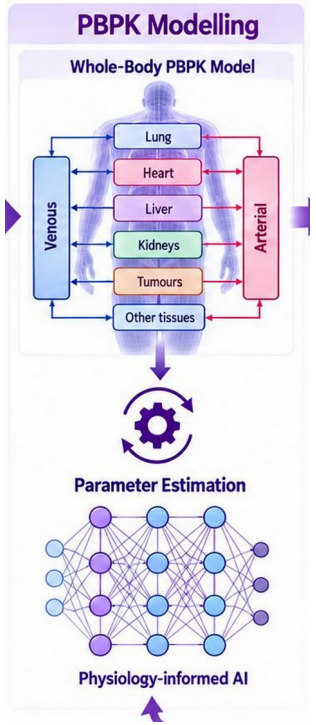
Longitudinal data keep updating the twin

THERANOSTIC DIGITAL TWINNING

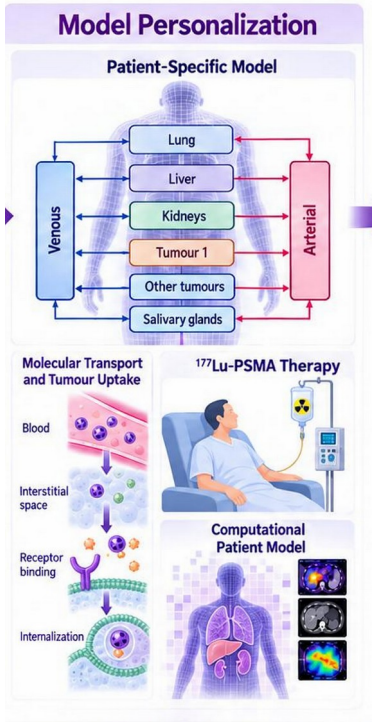
Building the twin: PBPK model for ^{177}Lu-PSMA



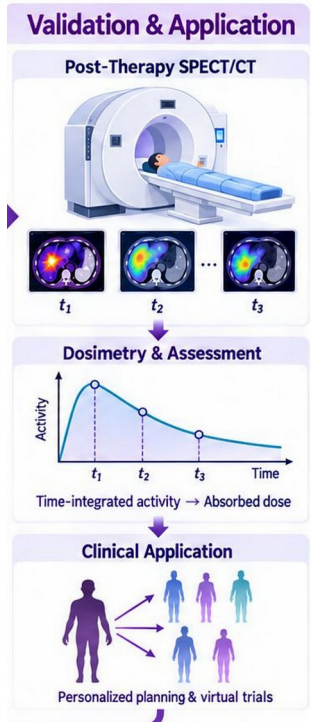
Patient data



PBPK model



Personalization

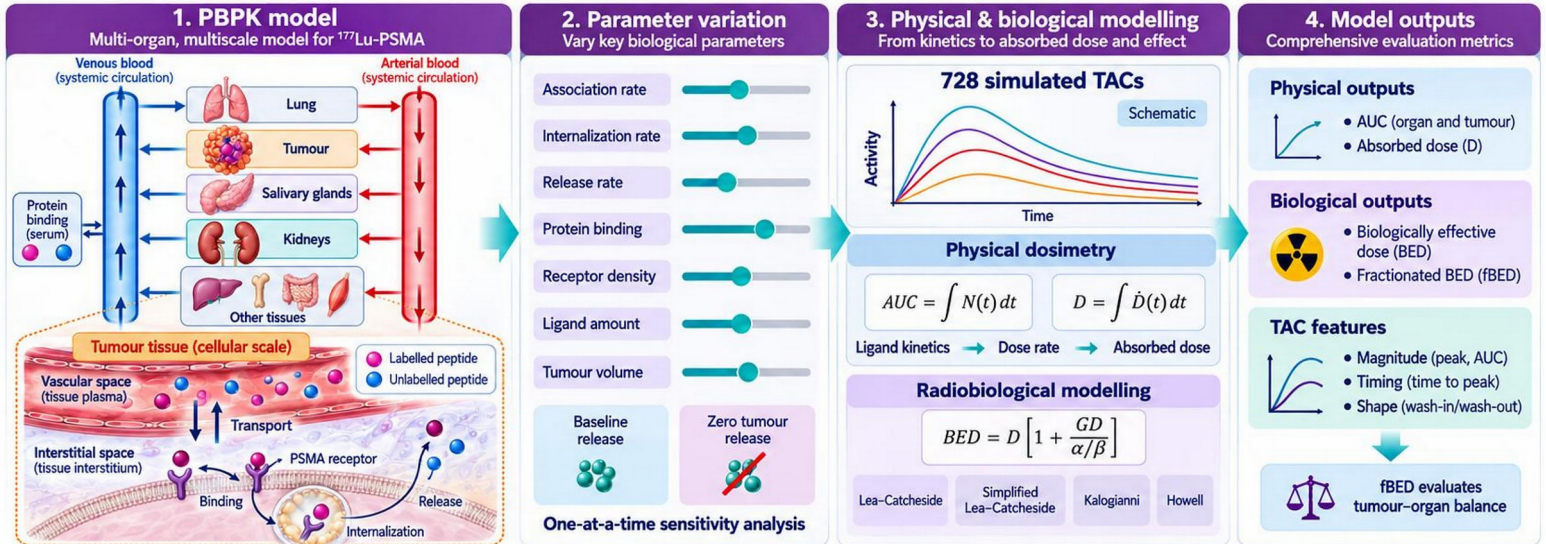


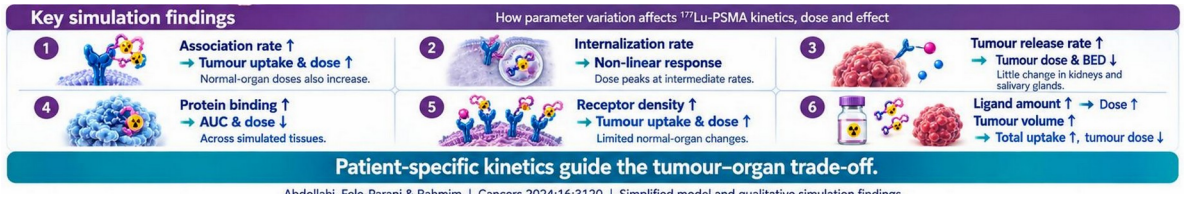
Validation & use

Model refinement with longitudinal data closes the loop

THERANOSTIC DIGITAL TWINNING

Model-informed ^{177}Lu-PSMA optimization

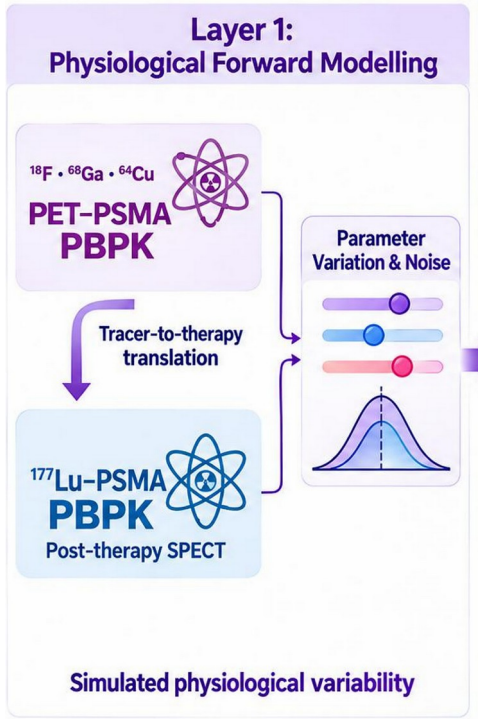




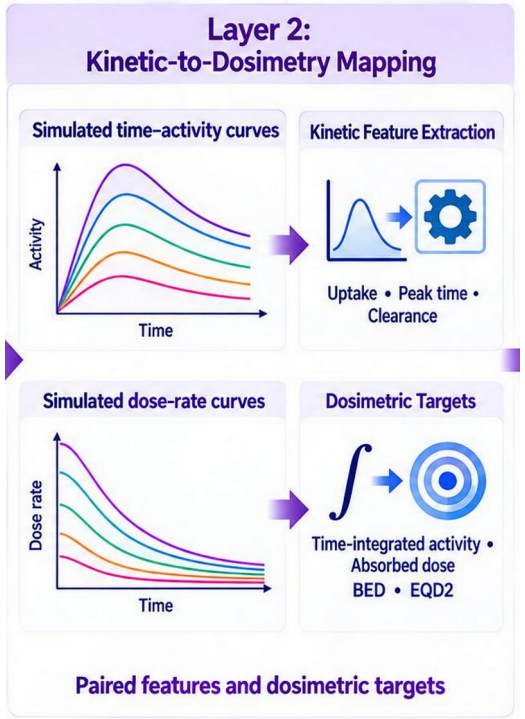
Patient-specific kinetics guide the tumour–organ trade-off

APPLICATION · PREDICTIVE DOSIMETRY

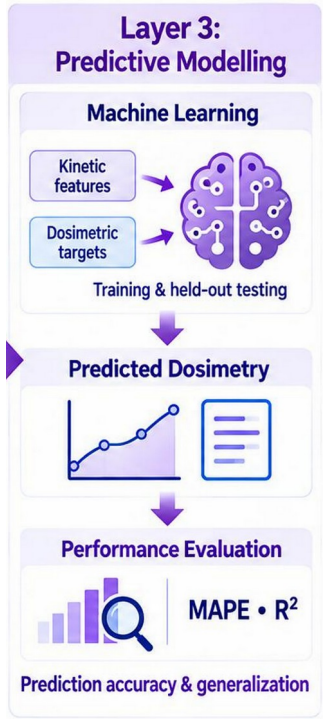
Predicting ^{177}Lu dosimetry from PET kinetics



Layer 1: physiological modelling



Layer 2: kinetics → dosimetry



Layer 3: machine learning

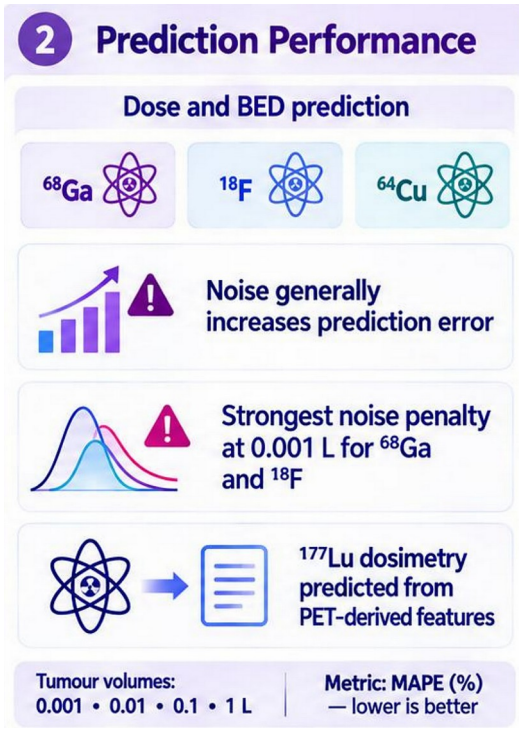
Pre-therapy PET kinetics → predicted therapy dose

APPLICATION · PREDICTIVE DOSIMETRY

Predictive dosimetry: key findings



Peak uptake (AMax) leads



Noise matters most in small tumours



Model interpretation (SHAP)

Uptake magnitude and kinetics support dosimetry prediction